



SOUTHEAST UNIVERSITY

Assignment on: Device Security Hardening and Static Routing
within a 4-Router WAN topology.
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Submitted by

MD. Kamran Hasan Rafin
Dept. of CSE
Batch - 60
ID : 2022000000066

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Experiment Name: Device Security Hardening and static Routing within a 4-Router WAN Topology.

Introduction: The deployment of robust network infrastructure requires not only the establishment of logical communication paths but also the implementation of stringent administrative security controls. In wide area network (WANs), where multiple routing devices interconnect geographically dispersed subnets, securing the CLI constitutes the primary defense against unauthorized administrative access.

This Lab experiments conducted using Cisco Packet Tracer. It explores two core competencies; 1. The configuration of device identity, access control mechanisms and legal warning banners for device hardening and 2. the manual configuration of static routing across a complex 4-Router square WAN topology to achieve complete network convergence. The experiment demonstrates how proper security configuration and precise routing logic together form a fully operational and secure enterprise network.

Objectives:

- Device Hardening: Configure hostnames, Banner MOTD, enable Passwords on all Routers.
- WAN Topology: Build a 4-router square ring using Serial DCE/DTE Interfaces with Class A IP address 10.0.0.0 to 17.0.0.0.
- Static Routing: Inject static routes on each router to achieve full inter-network reachability across 8 subnets.
- Verification: Validate security settings and routing convergence using show commands and temp Ping tests.

Theoretical Background:

1. Device Security Hardening: Network device Hardening is the process of securing a router or switch by reducing its attack surface.
- Hostname Configuration: The default hostname 'Router' provides no contextual or geographical information. In a multi-device environment, ambiguous hostnames are a leading cause of accidental misconfiguration. Assigning unique, descriptive hostnames [e.g: TOP-LEFT-ROUTER] is a fundamental network management

best Practice and is mandated in most enterprise Change management Policies.

- Banner MOTD: In enterprise and govt. architectures a legally Compliant MOTD (message of the day) banner is a regulatory requirement. It is displayed before the login Prompt and serves two critical functions. it explicitly warns unauthorized users that access is Prohibited and it legally establishes the owner's right to monitor and Prosecute unauthorized access. Without such a warning legal action against intruders may be compromised.
- Enable Password: The legacy 'enable Password' command stores Credentials in reversible. Password keeps ^{safe} our network from unauthorized logins.

2. WAN Serial Communication: In real world WAN, the Physical Clocking signal for serial links is Provided by the Carrier's equipment (CSU/DSU). In a simulated Packet Tracer environment one router must assume this Clocking responsibility. ~~The~~ The two roles in a serial link are:

- DCE (Data Communication Equipment): The router interface configured as the DCE endpoint is responsible for generating and Providing the Clocking signal for

the serial link. This is done using the clock rate [e.g: clock rate 9600]. In Packet Tracer, the DCE end is identifiable by the clock symbol on the cable.

- DTE (Data Terminal Equipment): The router DTE end of the serial link passively synchronizes to the clock signal provided by the DCE. No clock rate configuration is required on the DTE interface. The DTE simply processes incoming data according to the timing signal it receives.

Without a clock rate on the DCE end, the serial interface will remain in a 'down/down' state and no traffic can pass over that WAN link.

3. Static Routing Principles: Static routing involves an administrator manually programming forwarding entries into a router's routing table. Each static route entry specifies a destination network, its subnet mask and a next-hop IP address. Unlike dynamic routing protocols, static routing does not adapt automatically to topology changes but they offer deterministic, low-overhead forwarding suitable for simple or small topology.

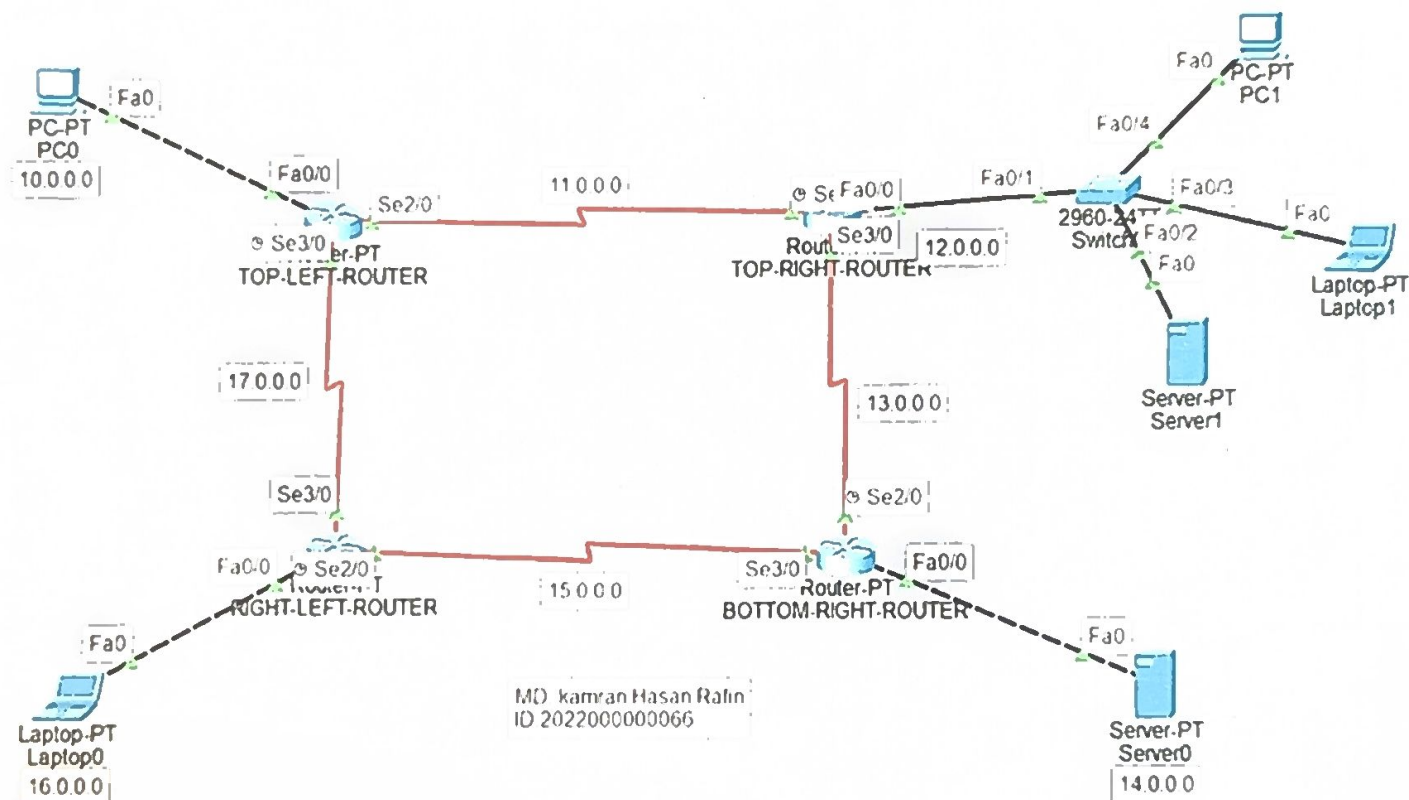
In a ring topology with 4 routers, each router must have static routes for all networks it does not directly own. With 8 total networks [4 LAN and 4 WAN] and each router owning 3, every router requires 5 static route entries to achieve full convergence.

□ Network Topology and Hardware Inventory: The simulated infrastructure was constructed in Cisco Packet Tracer as a square ring with four routers forming the WAN core and each router serving a distinct local Area network (LAN) segment at its edge. The topology provides two redundant paths between any two routers.

1. WAN Core Ring Networks: The four serial links forming the WAN ring are assigned dedicated Point-to-Point Class A Networks:

- 11.0.0.0 → Serial link between TOP-LEFT-Router and TOP-RIGHT-ROUTER.
- 13.0.0.0 → Serial link between TOP-RIGHT-ROUTER and BOTTOM-RIGHT-ROUTER.
- 15.0.0.0 → Serial link between BOTTOM-RIGHT-ROUTER and BOTTOM-LEFT-ROUTER.
- 17.0.0.0 → Serial link between BOTTOM-LEFT-ROUTE and TOP-LEFT-ROUTER.

SETTING HOSTNAME, BANNER AND PASSWORD



MO kamran Hasan Rafiq
ID 2022000000066

Time 24 38 18

Realtime Simulation



Scenario 0
New Delete
Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful		PC0	Laptop1	ICMP		0.000	N	0	(edit)	(delete)
Successful		PC0	Laptop0	ICMP		0.000	N	1	(edit)	(delete)
Successful		Laptop0	Server0	ICMP		0.000	N	2	(edit)	(delete)
Successful		PC1	Laptop0	ICMP		2.000	N	3	(edit)	(delete)

2. LAN segments: Each router connects to a unique LAN segment hosting the end-devices:

- 10.0.0.0 → managed by TOP-LEFT-ROUTER [PC0]
- 12.0.0.0 → managed by ~~BOTTOM~~ TOP-RIGHT-ROUTER [PC1, Laptop 1 via switch]
- 14.0.0.0 → managed by BOTTOM-RIGHT-ROUTER [server 0]
- 16.0.0.0 → managed by ~~RIGHT~~ BOTTOM-LEFT-ROUTER [Laptop 0]

Network Addressing Table:

Hostname	Interface	Network	IP Address	Subnet mask
TOP-LEFT-ROUTER	Fa0/0	10.0.0.0	10.0.0.1	255.0.0.0
TOP-LEFT-ROUTER	Se2/0	11.0.0.0	11.0.0.1	255.0.0.0
TOP-LEFT-ROUTER	Se3/0	17.0.0.0	17.0.0.1	255.0.0.0
TOP-RIGHT-ROUTER	Fa0/0	12.0.0.0	12.0.0.1	255.0.0.0
TOP-RIGHT-ROUTER	Se2/0	11.0.0.0	11.0.0.2	255.0.0.0
TOP-RIGHT-ROUTER	Se3/0	13.0.0.0	13.0.0.1	255.0.0.0
BOTTOM-RIGHT-ROUTER	Fa0/0	14.0.0.0	14.0.0.1	255.0.0.0
"	Se2/0	13.0.0.0	13.0.0.2	255.0.0.0
"	Se3/0	15.0.0.0	15.0.0.1	255.0.0.0
BOTTOM-left-Router	Fa0/0	16.0.0.0	16.0.0.1	255.0.0.0
"	Se2/0	15.0.0.0	15.0.0.2	255.0.0.0
"	Se3/0	17.0.0.0	17.0.0.1	255.0.0.0

☐ CLI Implementation and Configuration: Applied to all the four routers.

```
Router> enable
```

```
Router# Configuration terminal
```

```
Router(config)# hostname TOP-LEFT-ROUTER
```

```
TOP-LEFT-ROUTER(config)# banner motd '*****
```

```
Welcome
```

```
*****'
```

```
TOP-LEFT-ROUTER(config)# enable Password 1111
```

```
TOP-LEFT-ROUTER(config)# exit
```

☐ Static Route Injection: Each router requires 5 static routes to reach all non-directly connected networks.

→ Clockwise Path via se2/0:

- TOP-LEFT-ROUTER(config)# IP route 12.0.0.0 255.0.0.0 serial 2/0
- TOP-LEFT-ROUTER(config)# IP route 13.0.0.0 255.0.0.0 serial 2/0
- TOP-LEFT-ROUTER(config)# IP route 14.0.0.0 255.0.0.0 serial 2/0

→ Counter-Clockwise Path via se3/0:

- TOP-LEFT-ROUTER(config)# IP route 15.0.0.0 255.0.0.0 serial 3/0
- TOP-LEFT-ROUTER(config)# IP route 16.0.0.0 255.0.0.0 serial 3/0

[4] Verification Table: After ~~the~~ completing the hardening configuration ~~at~~^{its} time for icmp Ping were execution. Between all four network segments. All tests recorded Zero Packet loss.

Test Case	Source	Destination	Result	Latency
PC0 → Laptop1	10.0.0.1	12.0.0.1	Successful	0ms
PC0 → PC0	10.0.0.1	10.0.0.1	Successful	0ms
Laptop0 - Server0	16.0.0.1	14.0.0.1	Successful	0ms
PC1 → Laptop0	12.0.0.1	16.0.0.1	Successful	0ms

[4] Conclusion: The experiment successfully demonstrated device security hardening and static Routing across a 4-Router WAN topology. Hostname, Banner MOTD and enable Password Configuration secured all routers. While all the ~~to~~ devices manually injected static routes achieved complete network convergence. All icmp Ping tests confirmed zero packet loss between all network segments.